

Apriori Algorithm

▼ Objective :



Extract frequent itemset

```
[36]: from mlxtend.frequent_patterns import apriori
```

```
frequent_itemsets = apriori(basket , min_support = 0.02, max_len = 2, use_colnames = True, low_memory = True)
```

```
frequent_itemsets.head()
```

C:\Users\Hi\AppData\Roaming\Python\Python313\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames with non-bool types result in worse computational performance and their support might be discontinued in the future. Please use a DataFrame with bool type
warnings.warn(

```
[36]:
```

	support	itemsets
0	0.021613	(12 PENCIL SMALL TUBE WOODLAND)
1	0.021865	(12 PENCILS SMALL TUBE RED RETROSPOT)
2	0.021613	(12 PENCILS SMALL TUBE SKULL)
3	0.028607	(3 STRIPEY MICE FELTCRAFT)
4	0.028544	(4 TRADITIONAL SPINNING TOPS)

Despite reducing the minimum support threshold, only single-item frequent itemsets were identified. This indicates sparse co-purchase behavior due to a high number of unique products and low item overlap per transaction.

Association rule

```
[38]: from mlxtend.frequent_patterns import association_rules
```

```
Assoc_rules = association_rules(frequent_itemsets,  
                                metric="lift",  
                                min_threshold=1  
                                )
```

```
Assoc_rules
```

```
[38]:
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certain
0	(60 TEATIME FAIRY CAKE CASES)	(PACK OF 60 PINK PAISLEY CAKE CASES)	0.047952	0.050284	0.020290	0.423127	8.414828	1.0	0.017879	1.646319	0.925544	0.260307	0.3925
1	(PACK OF 60 PINK PAISLEY CAKE CASES)	(60 TEATIME FAIRY CAKE CASES)	0.050284	0.047952	0.020290	0.403509	8.414828	1.0	0.017879	1.596080	0.927816	0.260307	0.3734
2	(60 TEATIME FAIRY CAKE CASES)	(PACK OF 72 RETROSPOT CAKE CASES)	0.047952	0.075803	0.025898	0.540079	7.124731	1.0	0.022263	2.009467	0.902942	0.264649	0.5023

4	(ALARM CLOCK BAKELIKE GREEN)	(ALARM CLOCK BAKELIKE IVORY)	0.059924	0.034846	0.020668	0.344900	9.897947		1.0	0.018580	1.473293	0.956273	0.278912	0.3216
...	...	...	...	...	...	...	...		...	...	...	...	...	...
283	(WOODEN PICTURE FRAME WHITE FINISH)	(WHITE HANGING HEART T-LIGHT HOLDER)	0.068179	0.134279	0.025520	0.374307	2.787541		1.0	0.016365	1.383620	0.688180	0.144231	0.2776
284	(WOODEN PICTURE FRAME WHITE FINISH)	(WOODEN FRAME ANTIQUE WHITE)	0.068179	0.059798	0.033711	0.494455	8.268700		1.0	0.029634	1.859777	0.943381	0.357620	0.4623
285	(WOODEN FRAME ANTIQUE WHITE)	(WOODEN PICTURE FRAME WHITE FINISH)	0.059798	0.068179	0.033711	0.563751	8.268700		1.0	0.029634	2.135986	0.934972	0.357620	0.5318
286	(WOODEN STAR CHRISTMAS SCANDINAVIAN)	(WOODEN HEART CHRISTMAS SCANDINAVIAN)	0.030246	0.032262	0.023251	0.768750	23.828247		1.0	0.022276	4.184812	0.987913	0.592295	0.7610
287	(WOODEN HEART CHRISTMAS SCANDINAVIAN)	(WOODEN STAR CHRISTMAS SCANDINAVIAN)	0.032262	0.030246	0.023251	0.720703	23.828247		1.0	0.022276	3.472127	0.989972	0.592295	0.7115

"Association rules were filtered using confidence $\geq 30\%$ and lift ≥ 1.2 to retain only meaningful and business-relevant product associations."

Filtering rules: 1



```
[39]: rules_filtered = Assoc_rules[
      (Assoc_rules['confidence'] >= 0.6) &
      (Assoc_rules['lift'] >= 1.9)&
      (Assoc_rules['support'] >= 0.004)
      ]

rules_filtered = rules_filtered.sort_values(by = ["lift", "confidence"], ascending = False)

rules_filtered.head(10)
```

[39]:	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	c
273	(SET/6 RED SPOTTY PAPER PLATES)	(SET/6 RED SPOTTY PAPER CUPS)	0.032073	0.025331	0.020857	0.650295	25.672082	1.0	0.020045	2.787116	0.992892	0.570690	C
272	(SET/6 RED SPOTTY PAPER CUPS)	(SET/6 RED SPOTTY PAPER PLATES)	0.025331	0.032073	0.020857	0.823383	25.672082	1.0	0.020045	5.480375	0.986024	0.570690	C

▼ All 14 columns are not necessary these 5 are good enough [antecedents, consequents, support, confidence, lift]



```
[40]: Final_rule = rules_filtered[['antecedents', 'consequents', 'support', 'confidence', 'lift']]

Final_rule.head(2)
```

```
[40]:
```

	antecedents	consequents	support	confidence	lift
273	(SET/6 RED SPOTTY PAPER PLATES)	(SET/6 RED SPOTTY PAPER CUPS)	0.020857	0.650295	25.672082
272	(SET/6 RED SPOTTY PAPER CUPS)	(SET/6 RED SPOTTY PAPER PLATES)	0.020857	0.823383	25.672082

SET/6 RED SPOTTY PAPER PLATES) --> (SET/6 RED SPOTTY PAPER CUPS) & (SET/6 RED SPOTTY PAPER CUPS) --> (SET/6 RED SPOTTY PAPER PLATES) this both are considered as 1 association rule so we have to Clean our dataset

Cleaning

```
[41]: final_rules_updated = Final_rule.sort_values(
      by='confidence',
      ascending=False).drop_duplicates(subset=['support', 'lift'])
```

```
[48]: final_rules_updated.shape
```

```
[48]: (28, 5)
```

```
[48]: final_rules_updated.shape
```

```
[48]: (28, 5)
```

```
[47]: final_rules_updated
```

```
[47]:
```

	If customer buys	Then also buys	support	confidence	lift
41	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.037870	0.828966	13.604636
272	(SET/6 RED SPOTTY PAPER CUPS)	(SET/6 RED SPOTTY PAPER PLATES)	0.020857	0.823383	25.672082
252	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.035602	0.779310	12.367655
274	(SMALL MARSHMALLOWS PINK BOWL)	(SMALL DOLLY MIX DESIGN ORANGE BOWL)	0.021550	0.779043	24.006635
286	(WOODEN STAR CHRISTMAS SCANDINAVIAN)	(WOODEN HEART CHRISTMAS SCANDINAVIAN)	0.023251	0.768750	23.828247
44	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.045810	0.751810	11.931220
45	(ROSES REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.045810	0.727000	11.931220
39	(GARDENERS KNEELING PAD CUP OF TEA)	(GARDENERS KNEELING PAD KEEP CALM)	0.033711	0.722973	12.862759
19	(CHARLOTTE BAG PINK POLKADOT)	(RED RETROSPOT CHARLOTTE BAG)	0.032136	0.695771	10.976027

8	(ALARM CLOCK BAKELIKE GREEN)	(ALARM CLOCK BAKELIKE RED)	0.039509	0.659306	10.318724
121	(JUMBO BAG TOYS)	(JUMBO BAG RED RETROSPOT)	0.021298	0.652510	5.224686
273	(SET/6 RED SPOTTY PAPER PLATES)	(SET/6 RED SPOTTY PAPER CUPS)	0.020857	0.650295	25.672082
118	(JUMBO BAG STRAWBERRY)	(JUMBO BAG RED RETROSPOT)	0.031128	0.630102	5.045267
37	(DOLLY GIRL LUNCH BOX)	(SPACEBOY LUNCH BOX)	0.031821	0.624227	11.532584
63	(JUMBO BAG BAROQUE BLACK WHITE)	(JUMBO BAG RED RETROSPOT)	0.035539	0.623204	4.990037
116	(JUMBO BAG SPACEBOY DESIGN)	(JUMBO BAG RED RETROSPOT)	0.025898	0.618045	4.948727
126	(JUMBO BAG WOODLAND ANIMALS)	(JUMBO BAG RED RETROSPOT)	0.032892	0.617751	4.946375
47	(HAND WARMER SCOTTY DOG DESIGN)	(HAND WARMER OWL DESIGN)	0.020605	0.611215	15.203732
132	(JUMBO STORAGE BAG SUKI)	(JUMBO BAG RED RETROSPOT)	0.044865	0.606474	4.856073
32	(HOT WATER BOTTLE I AM SO POORLY)	(CHOCOLATE HOT WATER BOTTLE)	0.024008	0.601896	11.592337

Insights

- Customers frequently purchase similar or matching products together, especially within the same product category (like Customers buying paper cups are highly likely to also buy matching paper plates).
- Products with same color theme or design are strongly associated (like Customers tend to buy multiple color variants of the same product).
- Decorative and gifting items show very high lift values, even when support is moderate (WOODEN STAR CHRISTMAS → WOODEN HEART CHRISTMAS with Lift: 23.8)
- Most selected rules have Lift > 10, which means customers buying Product A are 10–25 times more likely to buy Product B than average.

Renaming Column

```
[44]: final_rules_updated.rename(columns={
      'antecedents': 'If customer buys',
      'consequents': 'Then also buys'},
      inplace = True)
```

```
[45]: final_rules_updated.head(10)
```

```
[45]:
```

	If customer buys	Then also buys	support	confidence	lift
41	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.037870	0.828966	13.604636
272	(SET/6 RED SPOTTY PAPER CUPS)	(SET/6 RED SPOTTY PAPER PLATES)	0.020857	0.823383	25.672082
252	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.035602	0.779310	12.367655
274	(SMALL MARSHMALLOWS PINK BOWL)	(SMALL DOLLY MIX DESIGN ORANGE BOWL)	0.021550	0.779043	24.006635
286	(WOODEN STAR CHRISTMAS SCANDINAVIAN)	(WOODEN HEART CHRISTMAS SCANDINAVIAN)	0.023251	0.768750	23.828247
44	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.045810	0.751810	11.931220
45	(ROSES REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.045810	0.727000	11.931220
39	(GARDENERS KNEELING PAD CUP OF TEA)	(GARDENERS KNEELING PAD KEEP CALM)	0.033711	0.722973	12.862759
19	(CHARLOTTE BAG PINK POLKADOT)	(RED RETROSPOT CHARLOTTE BAG)	0.032136	0.695771	10.976027
271	(SET/6 RED SPOTTY PAPER PLATES)	(SET/20 RED RETROSPOT PAPER NAPKINS)	0.022306	0.695481	11.581625

LIFT CHART(Bar chart for Association Rule vs Lift)

```
[55]: ## Top 10 rules according to lift

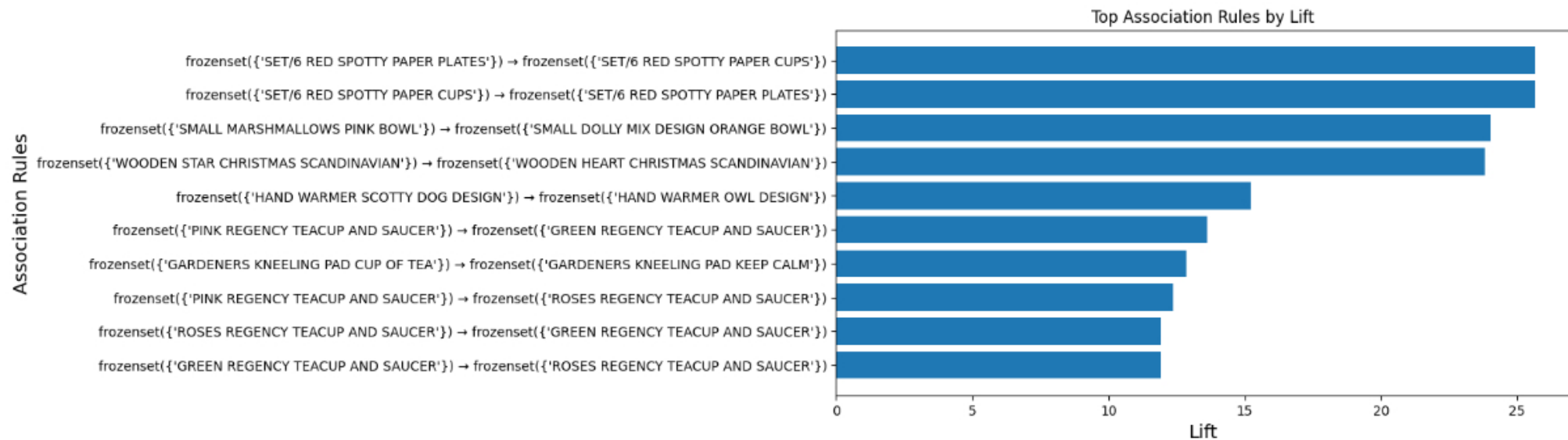
top_rules = final_rules_updated.sort_values(by = 'lift', ascending = False).head(10)

## Rule Label

top_rules['Rule'] = top_rules['If customer buys'].astype(str) + ' → ' + top_rules['Then also buys'].astype(str)
```

Horizontal Bar chart

```
[60]: plt.figure(figsize=(10,5))
plt.barh(top_rules['Rule'], top_rules['lift'])
plt.xlabel('Lift', fontsize = 14)
plt.ylabel('Association Rules', fontsize = 14)
plt.title('Top Association Rules by Lift')
plt.gca().invert_yaxis() ### Sorting
plt.show()
```



Insights

- Product pairs with high lift values indicate a strong likelihood of being purchased together compared to random chance.
- SET/6 Red Spotty Paper Plates and SET/6 Red Spotty Paper Cups show the highest lift, suggesting a strong complementary relationship.
- Such high-lift product combinations can be effectively used for bundle offers, cross-selling, and store layout optimization.
- Regency teacup and saucer variants show strong associations, indicating customer preference for matching product designs.